

# ED TRIAGE DESK

## Human-Computer Interaction (HCI) Decision-Support

### 1. PROBLEM SPACE

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| <p><b>PROBLEM STATEMENT</b></p> <p>ED triage nurses at Mercer General assess patients under a 3-minute decision window with incomplete information, while simultaneously recording vitals, managing a queue and answering patient questions. Existing triage decisions have no automated cross-check against presenting vitals and chief complaints. The triage desk interface surfaces a descriptive risk flag, not a numeric ESI recommendation, to prompt the nurse to review borderline cases, without replacing the decision.</p> |                                                                                                                                                                                                                                                                                                                                                                        |
| <p><b>USERS &amp; STAKEHOLDERS</b></p> <p><b>PRIMARY: TRIAGE NURSE</b></p> <p>Characteristics: Trained staff; 12h shifts; manages desk either alone or as a duo.</p> <p>Needs: Actionable in seconds, no added data entry.</p> <p><b>SECONDARY: NURSING DIRECTOR</b></p> <p>Characteristics: Oversight &amp; governance role.</p> <p>Needs: Proof of safety, auditability &amp; clinical judgement preservation.</p>                                                                                                                   | <p><b>GOALS</b></p> <p><b>PRIMARY USERS</b></p> <p>Short-term: Reduce inconsistent triage during high-volume encounters.</p> <p>Long-term: Build trust as a second check, not a replacement.</p> <p><b>SECONDARY USERS</b></p> <p>Short-term: Approve pilot with thought-through failure modes.</p> <p>Long-term: Reduce adverse triage events from acuity errors.</p> |

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| <p><b>2. ETHICAL CONSIDERATIONS</b></p> <p><b>AUTOMATION BIAS</b></p> <p>Risk of nurses deferring to the flag rather than exercising clinical judgement.</p> <p><b>ACCOUNTABILITY</b></p> <p>Clear ownership for missed escalations if a flag under-calls risk and the nurse does not override.</p> <p><b>DATA GOVERNANCE</b></p> <p>Establishing consent mechanisms at triage when patients arrive in acute distress.</p> | <p><b>3. GUIDELINES</b></p> <p><b>INTERACTION TYPE</b></p> <p>Passive decision-support only; nurse initiates every action; no auto-escalation.</p> <p><b>PATIENT/CLINICIAN INTERACTION</b></p> <p>Flag is nurse-facing only to prevent patient distress over uninterpretable flags.</p> <p><b>ETHICAL DESIGN</b></p> <p>Worded as prompt ('review vitals'), requires logged 1-line override reason.</p> |
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| <p><b>4. MINIMUM VIABLE PRODUCT</b></p> <p><b>CORE FUNCTION</b></p> <p>Display a descriptive risk flag (colour + icon + text) and top 3 contributing factors on triage desk screen within 1.5s of manual entry, with visible confirm/override control.</p> <p><b>OUT OF SCOPE</b></p> <p>EHR integration, device-stream vitals, audio alerts.</p> <p><b>SUCCESS METRIC</b></p> <p>Nurse override rate &lt; 15% after 2 weeks; zero adverse triage events attributable to the tool.</p> | <p><b>5 &amp; 6. ENVIRONMENT &amp; FORM</b></p> <p><b>PHYSICAL &amp; OPERATIONAL ENVIRONMENT</b></p> <ul style="list-style-type: none"><li>• Small, shared, cluttered desk; fluorescent glare; noise; power instability (load-shedding).</li><li>• Single primary user/shift; manual data entry; local network only.</li></ul> <p><b>FORM FACTOR &amp; INTERACTION</b></p> <ul style="list-style-type: none"><li>• Screen: Standard desktop/tablet display already in use.</li><li>• Input: Keyboard/touch screen matching desk setup.</li><li>• Output: Visual only (colour-coded flag + icon + label); no audio due to noise.</li></ul> |
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